

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is a Denial of Service Attack (DoS).

The logs show that the IP address (203.0.113.0) sends a lot of SYN packets to our web server (192.0.2.1) with a very fast pace. As a result, our server runs out of resources and cannot respond to other requests.

This event could be SYN Flooding, a type of Denial of Service attack that exploits the TCP three-way handshake.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol.

1. Synchronize (SYN): The client sends a SYN packet to the server asking to establish a connection.

2. Synchronize - Acknowledge (SYN-ACK): The server acknowledges the client's request, and requests to establish a connection with the client. Server reserves resources for their connection.

3. Acknowledge (ACK): The client acknowledges the server's request and the connection is fully open from now on.

When a malicious actor sends a large number of SYN packets all at once the server reserves resources for each request. A high volume of rapid, intentional connection requests can deplete server resources, exhausting system capacity and causing a Denial of Service (DoS) for legitimate users.

The logs indicate that our web server cannot process legitimate requests because it ran out of resources. The visitors receive error messages indicating that they cannot establish or maintain a connection to the web server. Server has stopped responding.